RIVER VALLEY HIGH SCHOOL
JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME							
CENTRE NUMBER	S						
CLASS	21J						
INDEX NUMBER							

BIOLOGY **9744/01**

Paper 1 Multiple Choice **23 September 2022**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

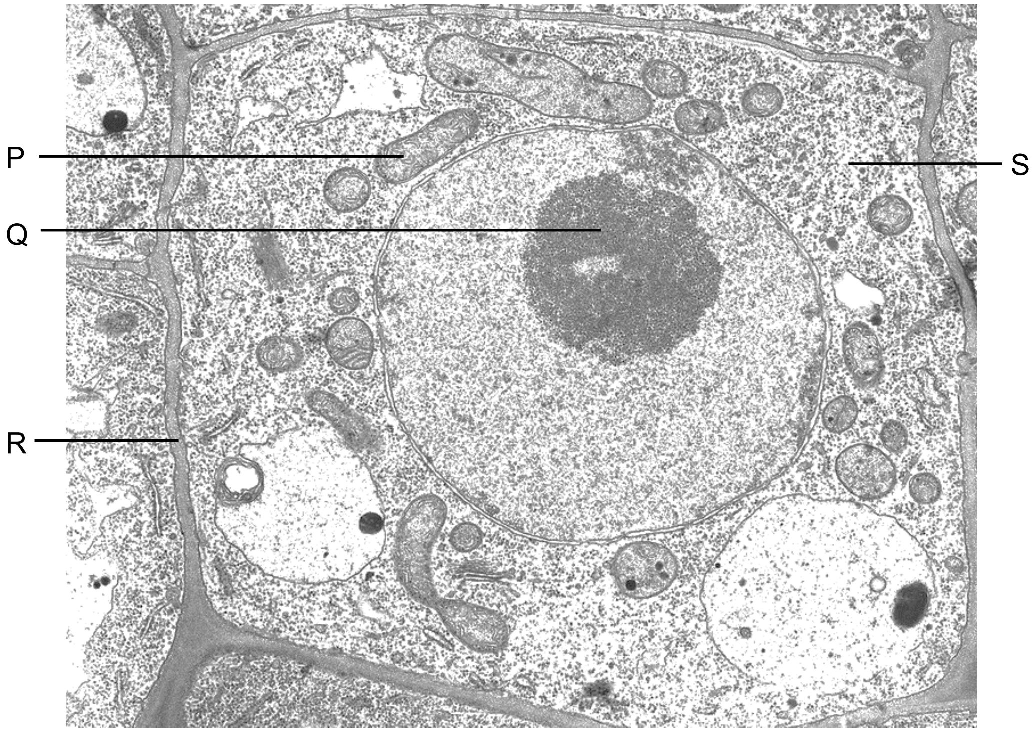
Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C**, and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

1 The electron micrograph shows part of several eukaryotic cells.



Which of the following statements are correct?

1 Both P and Q contain templates for transcription.

2 Q contains ribosomal RNA which is involved in peptide bond synthesis.

3 R has a fluid mosaic structure and regulates the movement of substances between the two cells.

4 S is the site of pyruvate synthesis.

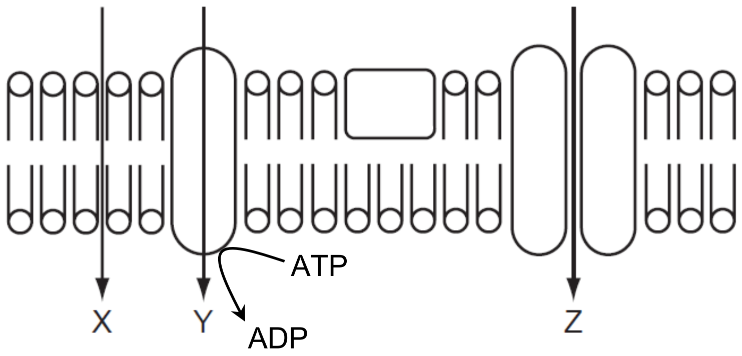
A 1, 2, 3 and 4

B 1, 2 and 4 only

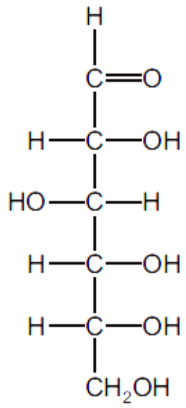
C 1 and 3 only

D 2 and 4 only

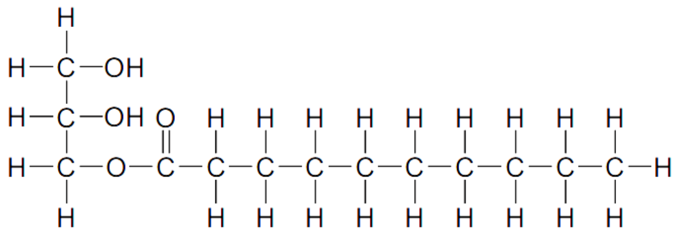
2 The diagram shows three pathways, X, Y and Z, through which substances can move across a cell surface membrane.



Which row correctly identifies all possible pathways through which substances 1, 2, 3 and 4 could use to enter the cell?



1



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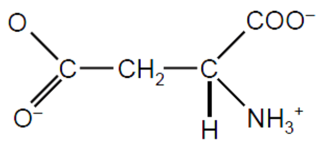

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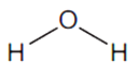
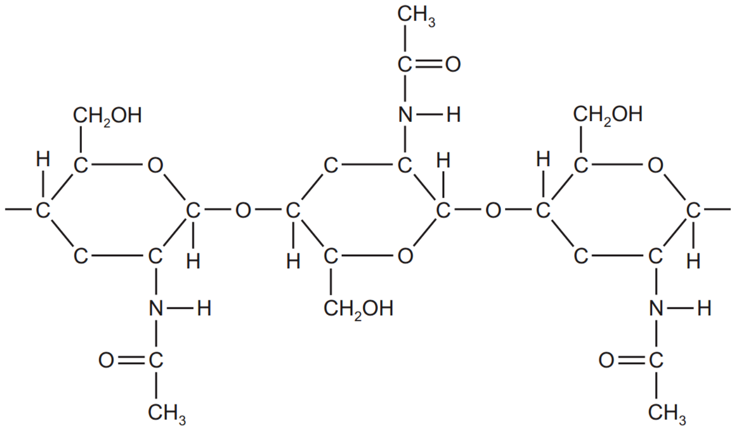


4

	X	Y	Z
A	2, 4	3	1, 3, 4
B	2, 4	1, 3	1, 3
C	2	1, 3	1, 3
D	2	3	1, 3, 4

3 The diagram shows the structure of part of a polysaccharide, chitin, found in the cell walls of certain fungi.



Which statements are true for **both** chitin and cellulose?

1 Each polysaccharide is composed of two different monosaccharides.

2 Each polysaccharide can form cross linkages via hydrogen bonds between chains.

3 Their monosaccharides are joined by 1,4-glycosidic bonds.

4 Every second monosaccharide in the polysaccharide chain is rotated by 180°.

A 1, 2 and 3

B 2, 3 and 4

C 2 and 4 only

D 3 and 4 only

4 Which row about the structure of proteins is correct?

	primary structure	secondary structure	quaternary structure
A	is the result of translation of an mRNA molecule by a ribosome into a chain of amino acids	occurs because of attraction between R groups of amino acid residues	is the sub-unit polypeptides that link together to form a protein
B	is the number of amino acids present in a protein	is the coiling of a chain of amino acids to form a β-pleated sheet or α-helix	contains two types of polypeptide that interact forming the shape of a protein
C	is synthesised by ribosomes in the cytoplasm	is the left-handed spiral formed by the primary structure	is formed by four polypeptides and an additional prosthetic group attached to the protein
D	is the sequence of amino acids in a protein coded by DNA	is formed by hydrogen bonding at intervals along the polypeptide backbone	is formed by the linking together of more than one polypeptide to form a protein

5 Hydrogenated vegetable oils are unsaturated fats that have been converted to saturated fats.

Which property of the fats will have changed?

1 Their melting point has increased.

2 There will be an increase in the ratio of carbon to hydrogen atoms.

3 Their hydrocarbon chains will fit together more closely.

A 1, 2 and 3

B 1 and 2 only

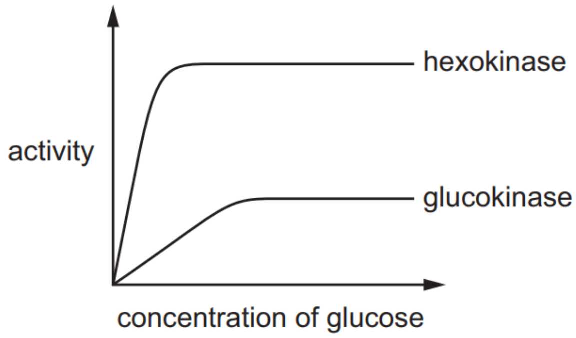
C 1 and 3 only

D 2 and 3 only

6 The enzymes glucokinase in the liver and hexokinase in the brain both catalyse the phosphorylation of glucose during glycolysis:

Glucose + ATP → glucose phosphate + ADP

The graph shows the activity of each enzyme measured at different concentrations of glucose.



Which statements correctly describe the differences between the two enzymes?

1 At low concentrations of glucose, temperature is limiting the activity of glucokinase only.

2 At high concentrations of glucose, the difference between the total amount of products formed by hexokinase and glucokinase remains constant.

3 Contact residues at the active site of hexokinase bind to glucose with greater affinity than that of glucokinase.

4 Hexokinase becomes saturated with glucose at a lower concentration of glucose than glucokinase.

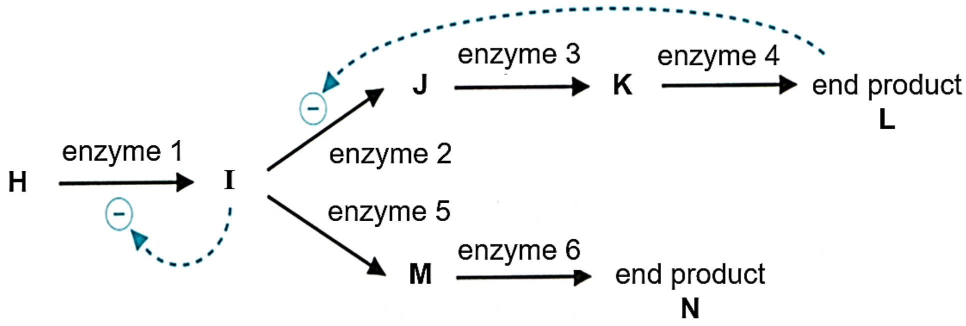
A 1, 3 and 4

B 2 and 3 only

C 2 and 4 only

D 3 and 4 only

7 The diagram shows a biosynthesis pathway.



The addition of substance **X** resulted in the following:

- no change in the concentration of metabolite **H**
- accumulation of metabolite **I**
- low concentrations of metabolite **J** and **K** and end product **N**

Further addition of metabolite **I** results in the formation of more end product **N**, but not end product **L**.

What does the information indicate about substance **X**?

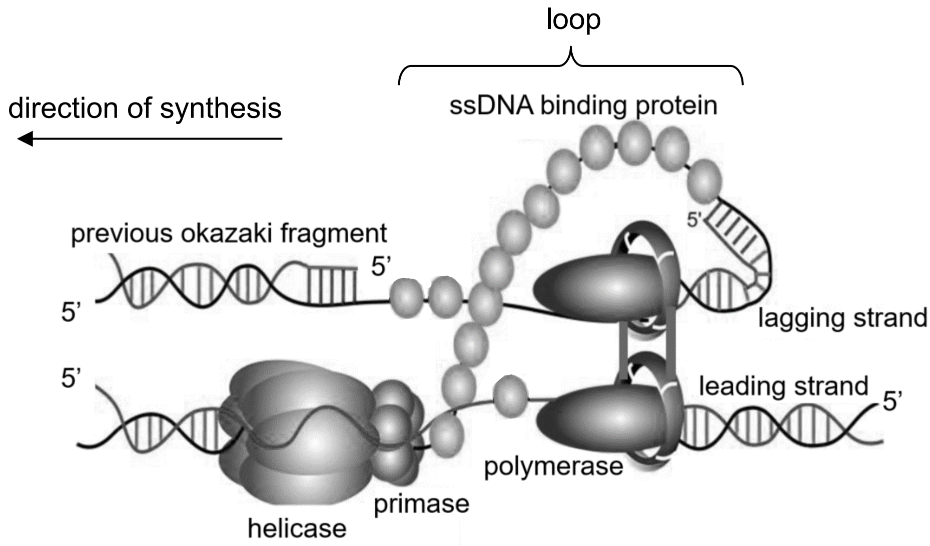
A It reacts with metabolite **J** to form metabolite **K**.

B It is an allosteric activator of enzyme 1.

C It is a non-competitive inhibitor of enzyme 3.

D It is a competitive inhibitor of enzyme 5.

8 The diagram shows the synthesis of the leading and lagging strands during DNA replication. The lagging strand template forms a loop at the replication fork.



Which statements explain why looping is needed during DNA replication?

1 The anti-parallel nature of the DNA strands.

2 DNA polymerase joins new nucleotides to the 3' end of the growing strand.

3 Single stranded DNA can be maintained to act as templates for polymerases.

4 Both DNA polymerases can then synthesise daughter strands in the same overall direction.

A 1, 2 and 3

B 1, 2 and 4

C 1 and 2 only

D 2 and 3 only

9 The DNA sequence CCAAGAAGTCGACAAACA was transcribed and translated to synthesise the polypeptide chain gly-ser-ser-ala-val-cys.

A mutation in the sequence resulted in a polypeptide chain that was shortened from six to two amino acids.

Which mutations might have occurred to result in the outcome observed?

1 A single base pair substitution from G to T.

2 A single base pair substitution from G to C.

3 Addition of a base pair at the 3^{rd} mRNA codon.

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

10 The following statements describe non-coding DNA in eukaryotes.

1 Presence of repeating sequences.

2 Number of nucleotides varies in different cells.

3 May be bound by proteins.

Which statement(s) apply **only** to non-coding DNA present outside of genes?

A 1 and 3 only

B 2 and 3 only

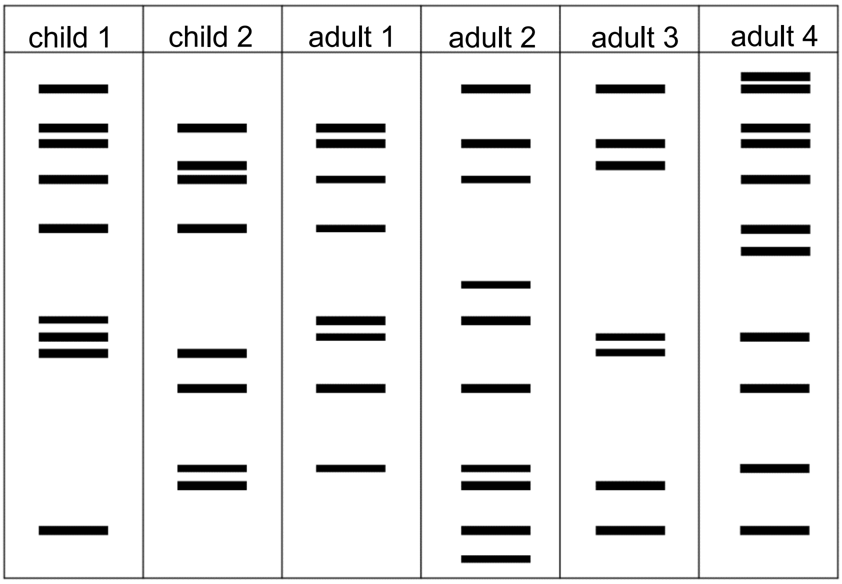
C 1 only

D 2 only

11 Which row describing the effect of a modification to the regulation of gene expression in eukaryotes is correct?

	modification	effect
A	mutation of intron splice sites	aggregation of misfolded proteins in cytoplasm
B	inactivation of poly-A-polymerase	accumulation of mRNAs in nucleus
C	loss-of-function mutation in eukaryotic initiation factors	inability to synthesise mRNA
D	over-expression of histone methyltransferase	upregulation of expression for a large variety of genes

12 DNA fingerprinting was used to identify the parents of two children. Short tandem repeats of a particular locus were selectively amplified using polymerase chain reaction and the products subsequently underwent gel electrophoresis. The results are shown below.



Given that both the children have the same pair of parents, which two adults may be their parents?

A adults 1 and 2

B adults 1 and 3

C adults 2 and 4

D adults 3 and 4

13 The mitotic cell cycle is carefully controlled by a cell-cycle control system.

Which of these statements explain why the mitotic cell cycle requires control?

1 To ensure sufficient time for DNA errors to be repaired.

2 To allow for genetic variation in daughter cells.

3 To induce apoptosis upon detection of any DNA damage.

4 To ensure that cells grow to an appropriate size for cell division.

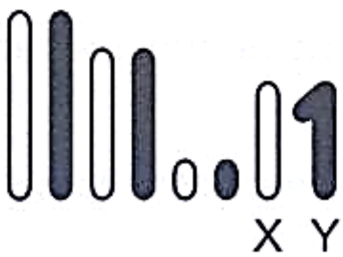
A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

14 The diagram shows the four pairs of homologous chromosomes present in the germ cell of a male fruit fly, *Drosophila melanogaster.*

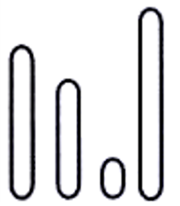


Given that crossing over does not occur during prophase I, which set of chromosomes shows the genetic variation resulting from independent assortment in the nucleus of a sperm cell?

A

B

C



D

15 The Philadelphia chromosome is associated with the development of several forms of leukaemia. The Philadelphia chromosome is formed when:

- one end of chromosome 9 and one end of chromosome 22 breaks
- part of the *BCR* gene that lies at broken end of chromosome 22 fuses with the *ABL1* gene that is part of the fragment of chromosome 9

When the *BCR-ABL1* fused gene is expressed, a protein that exhibits high tyrosine kinase activity is produced.

Which statement(s) cannot be concluded with the information above?

1 *BCR-ABL1* gene is a proto-oncogene as the gene product stimulates cell division.

2 Multiple copies of *BCR* gene results in the development of cancer.

3 The Philadelphia chromosome is an example of chromosomal translocation.

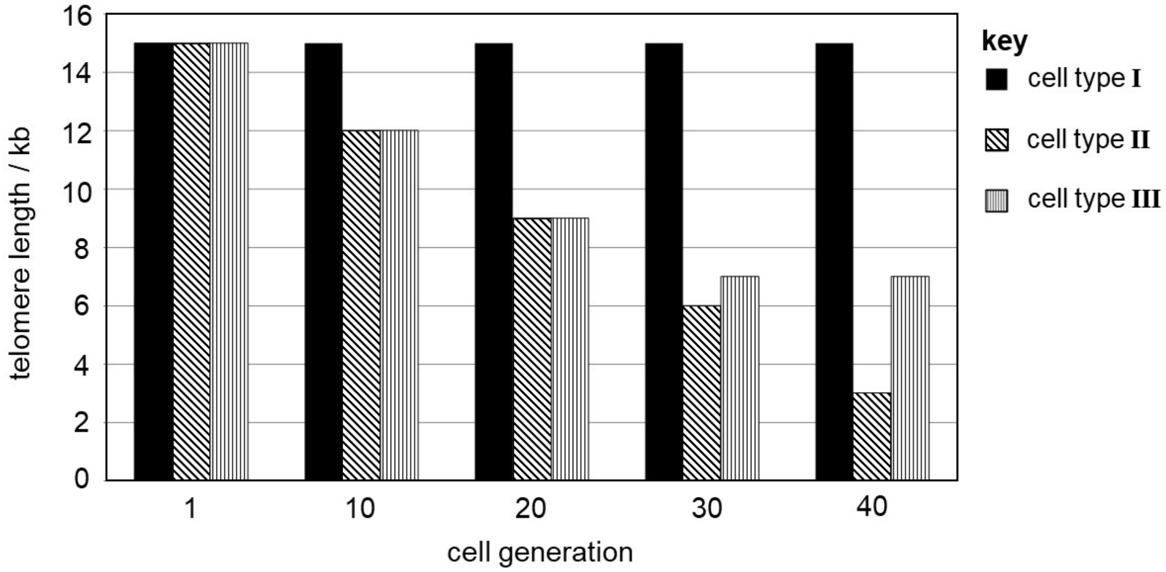
A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 only

16 The graph shows the telomere length of three different cell types **I** to **III**.



What may be concluded from the graph above?

1 Cells of type **I** and **III** are capable of long term self-renewal.

2 Cells of type **II** do not contain mutations in tumour suppressor genes.

3 Cells of type **I** can be induced to differentiate by environmental signals.

4 Cells of type **III** can proliferate uncontrollably.

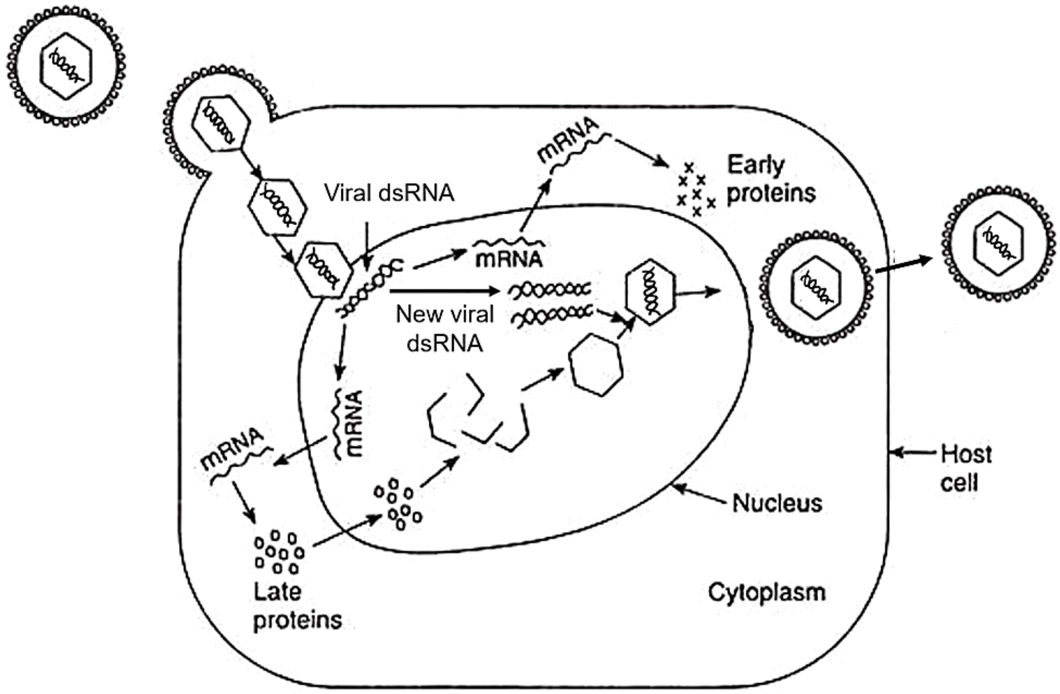
A 1, 2 and 3

B 1, 3 and 4

C 1 and 3 only

D 2 and 4 only

17 The diagram shows the reproductive cycle of a virus in a host cell.



Which statement correctly describes the reproductive cycle of this virus?

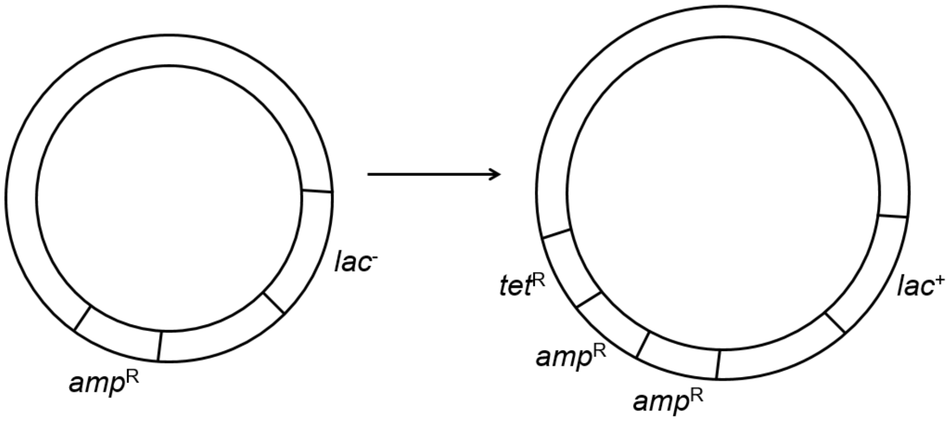
A It is an enveloped virus that enters the host cell via receptor-mediated endocytosis.

B The viral genes serve as templates to form RNA molecules with negative configuration to synthesise early and late proteins.

C Synthesis of new viral genome is carried out by the same enzyme as in the influenza virus.

D The viral progeny is released via budding.

18 Which environment could account for the changes in the bacterial chromosome as shown in the diagram?



key

amp^R – ampicillin resistance gene

tet^R – tetracycline resistance gene

lac$^+$ – lac operon present

lac$^-$ – lac operon absent

A Exposure of actively dividing bacteria to mutagens.

B A single bacterial colony grown in culture medium containing lambda phages.

C Bacteria grown in culture medium with other strains of heat-killed bacteria.

D Bacteria grown in culture medium containing tetracycline.

19 The activities of the *lac* operon in four different *E. coli* strains, cultured in different nutrient media, were investigated. The results are shown in the table below.

E. coli strain	nutrient present in media	conformational state of *lac* repressor	β-galactosidase
1 (normal)	glucose	active	absent
	lactose	inactive	present
2 (mutant)	glucose	active	present
3 (mutant)	lactose	active	absent
4 (mutant)	lactose	inactive	absent

Which statements could explain these results?

1 A mutation in the *lac I* gene in strain 3.

2 A mutation in the operator of the *lac* operon in strain 2.

3 A mutation in the promoter of the *lac* operon in strain 4.

4 A mutation that swapped the positions of the promoter and operator in the *lac* operon in strain 2.

A 1, 2, 3 and 4

B 1, 2 and 3 only

C 1 and 2 only

D 3 and 4 only

20 Coat colour in domestic cats is controlled by two genes. Information of these genes are given in the table below.

gene location	symbol	phenotype
chromosome 2	**B**	black coat colour
	b	brown coat colour
X chromosome	$\mathbf{X^G}$	ginger coat colour
	$\mathbf{X^g}$	absence of ginger colour $\mathbf{X^G}$ is expressed even in the presence of the **B** allele

During the development of female cats, one X chromosome in each cell is inactivated at random. Consequently, in female cats that are heterozygous at both gene loci, skin cells may be in a patch of ginger fur or black fur depending on whether they have developed from a cell with an active $\mathbf{X^G}$ allele or $\mathbf{X^g}$ allele. Cats with this form of colouring are called tortoiseshell cats.

A male cat heterozygous for black coat is mated with a female cat with tortoiseshell coat.

What proportion of their kittens would be expected to have the tortoiseshell colouring?

A 0.125

B 0.25

C 0.5

D 0.75

21 In domestic poultry, the gene for white feathers is dominant to the gene for dark feathers and the gene for frizzled feathers is dominant to normal feathers. Pure-breeding poultry which were homozygous for dark, frizzled feathers were crossed with pure-breeding poultry bearing white, normal feathers. The F1 were test-crossed and the phenotypes of offspring were recorded. The results are shown.

white frizzled	39
dark frizzled	79
white normal	68
dark normal	46

The chi-squared test can be used to determine the probability that offspring ratio fits the expected ratio.

The formula for the chi-squared test is:

$$\chi^2 = \Sigma \frac{(O-E)^2}{E} \qquad v = c - 1$$

Part of the probability table for chi-squared values is shown below.

degrees of freedom	probability, p			
	0.10	0.05	0.01	0.001
1	2.71	3.84	6.64	10.83
2	4.61	5.99	9.21	13.82
3	6.25	7.82	11.35	16.27
4	7.78	9.49	13.28	18.47

Which combination correctly describes the conclusion of the χ^2 test?

	probability	loci of genes
A	< 0.05	different chromosomes
B	< 0.05	same chromosomes
C	> 0.05	different chromosome
D	> 0.05	same chromosome

22 Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?

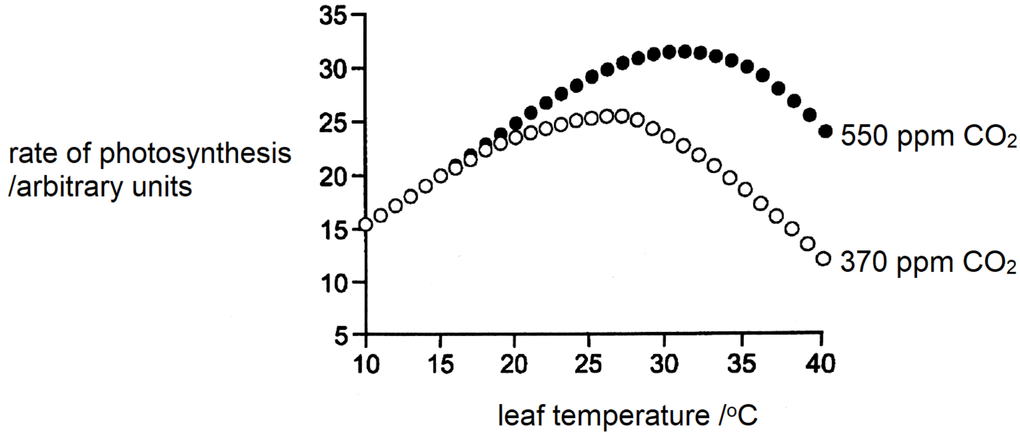
A Identical genetic crosses performed between varieties of *Dendranthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glass house.

B Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.

C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.

D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

23 The graph shows the results of increased concentration of carbon dioxide on soy bean photosynthesis at various leaf temperatures. Carbon dioxide concentration is measured in ppm (parts per million). Light intensity was at an optimum level.



Which statement(s) concerning the data in the graph is valid?

1 When temperature is limiting, increased carbon dioxide concentration increases the rate of photosynthesis.

2 At all temperatures up to 27.5 °C, temperature is the only limiting factor for both carbon dioxide concentrations.

3 At all temperatures, carbon dioxide concentration is the main limiting factor.

4 The photosynthetic rate obtained at the optimum temperature for 370 ppm CO_2 could be achieved at a temperature around 7.5 °C lower using an increased concentration of 550 ppm CO_2.

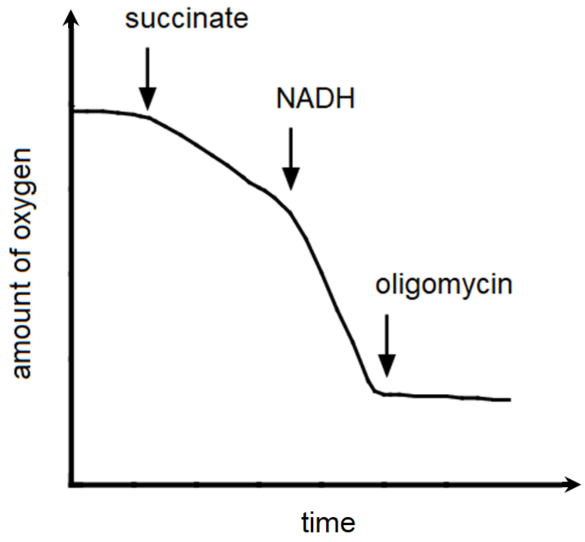
A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 4 only

24 A suspension of mitochondria was prepared in a buffer containing ADP and inorganic phosphate (Pi). The oxygen concentration in the buffer was monitored carefully and recorded as shown below. At the times indicated, a specific reagent was added to the buffer. Throughout the experiment, the concentrations of ADP and Pi were in excess.



Which one of the following shows correctly from the highest to the lowest, the rate of ATP production after the addition of the three chemicals?

	highest rate ⟶		lowest rate
A	succinate	NADH	oligomycin
B	NADH	succinate	oligomycin
C	succinate	oligomycin	NADH
D	oligomycin	NADH	succinate

25 The steps involved in a cell-signalling pathway are listed.

- binding of ligand to receptor binding site
- phosphorylation of receptor
- synthesis of second messengers
- activation of the enzyme adenyl cyclase
- initiation of a protein kinase cascade
- binding of protein to DNA sequences

What is the 3rd common step that occurs during **both** insulin and glucagon signalling?

A synthesis of second messengers

B activation of the enzyme adenyl cyclase

C initiation of a protein kinase cascade

D binding of protein to DNA sequence

26 Before the settlement of California in the 1800s, the elk population was very large. By about 1900, there were only a few dozen elk left due to hunting.

Owing to protection, there are now about 3000 elk living in a small number of isolated herds.

Unfortunately, some of the elk in all the herds have difficulty grazing due to a shortened lower jaw, as a result of having one copy of a mutated allele. These elk can be observed in subsequent generations.

Which statements best explain the observations?

1 There was a mutation affecting jaw size in some of the herds.
2 There was random mating within each herd.
3 The mutated allele is preserved in the population due to heterozygote protection.
4 The current elk distribution demonstrates a founder effect.

A 1 and 2

B 1 and 3

C 2 and 4

D 3 and 4

27 Lord Howe Island is relatively small, volcanic island that formed approximately 6.4 – 6.9 million years ago. Two species of palm, *H. forsteriana* and *H. belmoreana*, are found on this island and had descended from one ancestor species.

To investigate speciation of the two palm species, the following findings are listed.

1 Two populations from the ancestral species develop variation in soil tolerance. One population grows in neutral to acidic soil while the other grows in soils rich in calcarenite which has a more basic pH.

2 Calcarenite soils are poor in nutrients which will delay the physiological development of palm.

3 The soils on most parts of Lord Howe Island are mainly neutral to acidic, except at low lying areas where alkaline soils are found.

4 The peak flowering of each species is separated by approximately six weeks and has limited overlap.

What is the correct sequence to explain this speciation?

A 1 → 2 → 3 → 4

B 2 → 4 → 3 → 1

C 3 → 1 → 2 → 4

D 4 → 2 → 1 → 3

28 Which statement about immunity is correct?

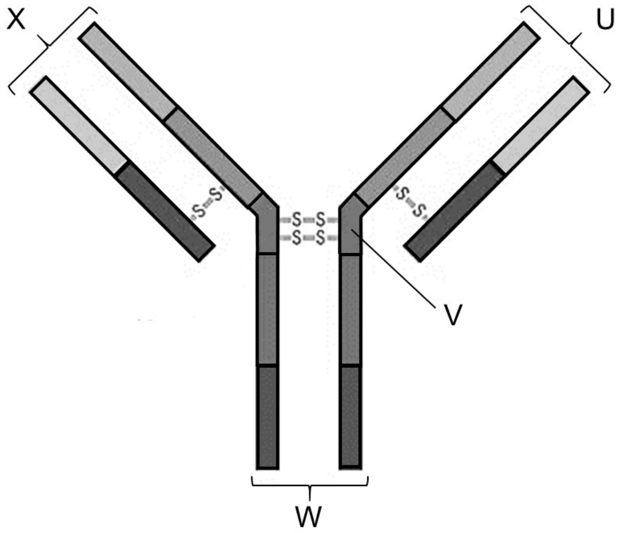
A Immunity following a vaccination lasts for a greater length of time than natural passive immunity.

B Antibody injection results in artificial active and artificial passive immunity.

C Natural active immunity provides a faster response to infection than artificial active immunity.

D Antigen presentation by specific B lymphocytes only occurs with natural immunity.

29 The diagram shows the simplified structure of an antibody.



Which statement is correct?

A U and X are unique antigen-binding sites that allow the antibody to bind to two different antigens.

B V allows for flexibility of the antibody to bind to antigens.

C W varies across different classes of antibodies due to alternative splicing of heavy chain mRNA.

D X undergoes somatic hypermutation during B lymphocyte development in bone marrow.

30 Dengue disease is spread by *Aedes aegypti* and *Aedes albopictus* mosquitoes.

Which of these risk factors increases the probability of developing severe dengue disease following infection?

A increased resistance of the vector to insecticides

B exposure to different mosquito species

C people living in more crowded conditions due to an increase in human populations

D having a chronic disease such as diabetes